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Faculty of Engineering and Architecture
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Digital Traffic Light System Project Report

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Course: MEE214 - Digital Electronics

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1. Project Summary and Scenario Definition

The objective of this project is to design a fault-tolerant digital traffic light control unit that safely coordinates vehicle and pedestrian traffic. The designed system manages three primary flows: the main road, the side road, and the pedestrian crossing. These phases, referred to as 'states' in digital electronics, do not represent random output combinations but rather meaningful and safe traffic flows. In this sensorless system, external uncertainties are represented by manual switches (pedestrian button), and the effects of these inputs on the system are controlled at the hardware level.

2. System Architecture and Hardware Design

The designed circuit operates through the integration of sequential and combinational logic circuits. The electronic components used in the physical implementation of the system, the circuit schematic, and the functional blocks are detailed below.

2.1. Bill of Materials (BOM)

The hardware components used in the Proteus design of the circuit are categorized by their functional groups as follows:

Integrated Circuits (Logic and Timing):

- 1x NE555 (Timer IC)
- 1x 74HC74 (Dual D-Type Flip-Flop - Sequential Logic)
- 1x 74HC86 (XOR Gate)
- 2x 74HC32 (OR Gate - U2, U9)
- 2x 74HC08 (AND Gate - U5, U7)
- 1x 74HC00 (NAND Gate - U8)
- 1x 74HC04 (Inverter / NOT Gate - U6)

Semiconductors (Diodes and Transistors):

- 2x BC237 NPN Transistors (Q1, Q2)
- 4x 1N4148 Fast Switching Diodes (D1, D2, D10, D11)

Display Elements (LEDs):

- 3x Green LEDs (Main Road, Side Road, Pedestrian)
- 3x Red LEDs (Main Road, Side Road, Pedestrian)
- 1x Yellow LED (Main Road)

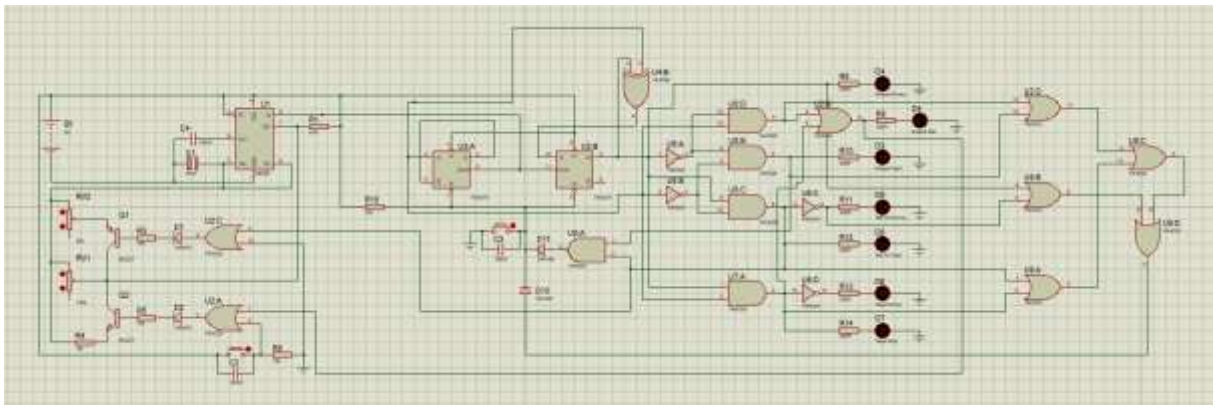
Passive Components (Resistors and Capacitors):

- 1x 100 k Ω Potentiometer (RV1)
- 1x 50 k Ω Potentiometer (RV2)
- 1x 47 k Ω Resistor (R1)
- 6x 10 k Ω Resistors (R3, R4, R5, R6, R15, and general use)
- 7x 330 Ω Resistors (LED current limiters, R8-R14)
- 1x 44 μ F Electrolytic Capacitor (C1)
- 3x 100 nF Ceramic Capacitors (C2, C3, C4)

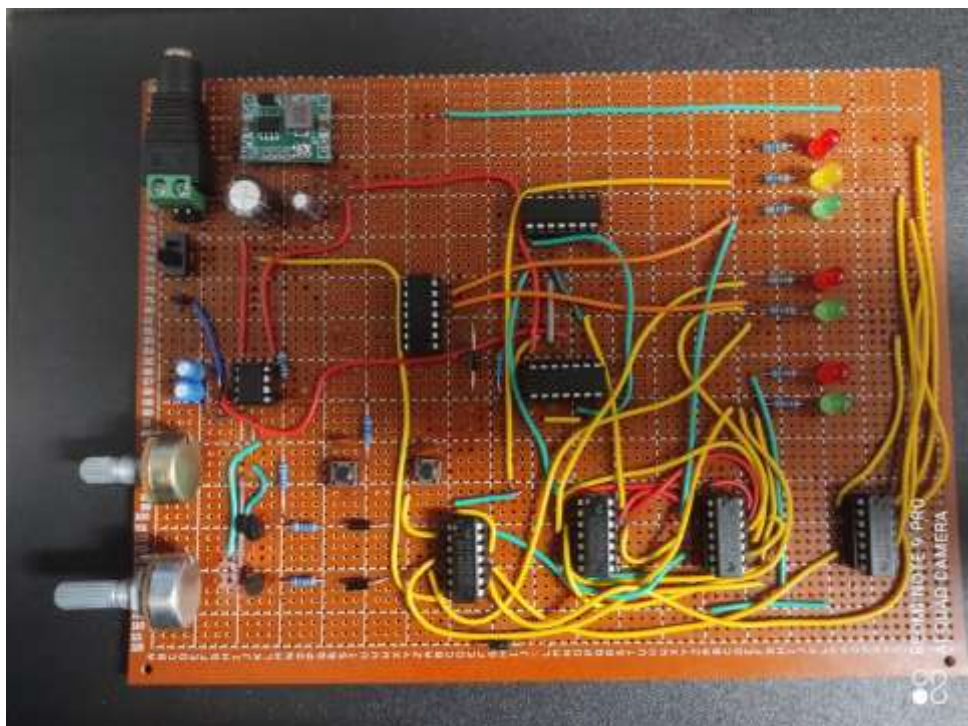
Power Supply:

- 1x 5V DC Battery/Power Source (B1)

2.2. Circuit Schematic



2.3. Physical Implementation



2.4. Functional Blocks

The circuit operates through four main functional blocks using the specified hardware components:

- **Clock Pulse Generation:** The NE555 timer IC is utilized in "astable" mode for timing. The generated square wave signal ensures the synchronization of the entire sequential logic. Transistors Q1 and Q2, along with potentiometers RV1 and RV2, allow the dynamic adjustment of the system's speed by altering the RC time constant.
- **State Machine (Sequential Logic):** Two 74HC74 D-type Flip-Flops are interconnected with a 74HC86 XOR gate to form a 2-bit synchronous counter. This structure represents the core state machine that determines the sequence in which the traffic lights will cycle.
- **State Decoder (Combinational Logic):** The flip-flop outputs (QA, QB) are decoded via 74HC08 (AND) and 74HC04 (Inverter) gates. Certain light combinations must strictly be prevented. This logic block performs the hardware interlocking, averting hazardous situations such as the main road and side road having green lights simultaneously.
- **Outputs and Indicators:** The decoded logic signals drive 7 LEDs through 330 Ω current-limiting resistors (Main Road: Green/Red/Yellow, Side Road: Green/Red, Pedestrian: Green/Red).

3. State Analysis and Traffic Cycle

The system transitions sequentially and safely among the four main states outlined below. Cross-traffic flow is not permitted at any stage of the transitions.

Phase (State)	Main Road Lights	Side Road Lights	Pedestrian Lights	Description
State 0	GREEN	RED	RED	Traffic flows on the main road.
State 1	YELLOW	RED	RED	The main road prepares to stop.

Phase (State)	Main Road Lights	Side Road Lights	Pedestrian Lights	Description
State 2	RED	GREEN	RED	Right of way is given to the side road.
State 3	RED + YELLOW	RED	GREEN	Safe crossing duration for pedestrians.

4. Advanced Features and Error Management (Fail-Safe)

To ensure hardware robustness in traffic control systems, specific error management and asynchronous intervention features have been incorporated into the circuit.

Hardware logic gates guarantee that the system consistently returns to a known, safe state in the event of unexpected or illegal states.

- Asynchronous Pedestrian Intervention and Cycle Acceleration:** When a pedestrian request is received, the system reacts immediately yet safely. Upon pressing the pedestrian button, an asynchronous intervention is triggered via the U2 (74HC32) OR gates, which are integrated with debounced button and capacitor networks (C2, C3). As long as the pedestrian holds the button, a 10 kΩ resistor, activated through the U2:A gate, is introduced into the timing circuit. This intervention alters the RC charging time of the 555 timer, accelerating the state machine cycle and enabling the system to reach the pedestrian crossing phase more rapidly but in a sequential manner.
- Hardware Auto-Reset and Conflict Protection:** To prevent the system from locking up or entering a hazardous phase, two distinct hardware "fail-safe" scenarios have been designed:
 - Dual Green (Critical Error) Protection:** The simultaneous green lighting of both the main road and the side road poses the most critical traffic hazard. In the event of such a failure due to a logic error, the U8:A (74HC00 NAND) gate instantly detects this logical conflict. Upon detection, the system is automatically reset, and the intersection is immediately forced into its safe initial state (State 0 - Main Road Green, Side Road Red).
 - Dead State (Dark Intersection) Protection:** A dedicated supervisory network has been established against the possibility of the system

entering an invalid state where no lights are illuminated (all outputs being LOW) during power-up or operation. This detection block, consisting of U2:D and U9:A, B, C, D (74HC32 OR) gates, continuously monitors the status of all lights in the system. When it detects that no lights are on, these gates reset the system, restarting the cycle from the safe initial state.

5. Frequency and Timing Analysis

The operating frequency of the system varies according to external factors (button and transistor triggers). The timing data obtained under different resistance combinations are as follows:

Operating Mode	Equivalent Resistance	HIGH Duration	LOW Duration	Total Period	Frequency Range
1st State (Transistors OFF)	100 k Ω - 0 Ω	4.48 s	3.05 s	7.53 s	0.133 Hz - 0.699 Hz
2nd State (Q1 ON)	33.33 k Ω - 0 Ω	2.45 s	1.02 s	3.47 s	0.288 Hz - 0.699 Hz
3rd State (Q2 ON)	9.09 k Ω - 0 Ω	1.71 s	0.28 s	1.99 s	0.503 Hz - 0.699 Hz

(Analysis Inference: As the potentiometer values approach the 0 Ω level, the frequencies in all modes converge and equalize at the upper limit of 0.699 Hz. To accurately observe the acceleration/deceleration stages, maintaining the potentiometers at medium or high resistance settings provides the optimum operating range.)

6. Conclusion and Evaluation

In this project, fundamental digital logic concepts were successfully transformed into a physical, time-aware, and safety-critical control system. The final output of the project is not merely a simple timer that drives LEDs at specific intervals; it is a robust traffic light architecture that supervises state transitions, incorporates hardware interlocking, and can automatically revert to a safe state (fail-safe) in unexpected situations.

During the design process, an NE555-based clock pulse generator, D-Type flip-flops, and 74 series logic gates were integrated, thereby resolving the system's timing and sequential logic requirements entirely at the hardware level, without the use of any microcontroller. The integration of asynchronous inputs from the external world, such as a pedestrian button, into the main cycle without compromising system safety (e.g., cross-traffic conflict) was successfully analyzed and implemented.

Consequently, this study has enabled the anticipation of potential risks in an intersection scenario (such as the simultaneous green lighting of both roads or the extinguishing of all lights) and has ensured that these risks are not merely eliminated as software probabilities, but rendered hardware-impossible. The consistent and safe operation of the design on physical hardware, extending beyond theoretical analyses and simulations, demonstrates the robust engineering approach and reasoning required for the construction of real-world control systems.